Workshop 3.2: Emerging Technologies (AI/ML/Blockchain/IoT)

Phase 3: Digital Innovation

Workshop Code: WS 3.2
Primary Component: Data Centre Expansion (Tech Hub #2-5 Combined)
Duration: 2.5 days (4-day span with hands-on time)
Timing: Month 8, Week 2-3
Phase Theme: "Empowering the Digital Ecosystem"

Executive Summary

This intensive workshop explores emerging technologies—Artificial Intelligence/Machine Learning, Blockchain, and Internet of Things—and their applications in government and the Caribbean context. Using the sandbox environments established in Workshop 3.1, participants will learn about, experiment with, and design pilots for these technologies. This workshop consolidates what the TOR envisioned as multiple tech hub workshops (#2-5) into a single comprehensive experience, recognizing that these technologies often interconnect and that a holistic understanding creates the most value.

Key Outcome: Emerging technology strategy with concrete pilot projects ready for implementation in specialized sandboxes.

Workshop Objectives

Primary Objectives

1. **Build Capacity:** Demystify AI/ML, blockchain, and IoT for government stakeholders
2. **Identify Applications:** Discover practical government use cases for emerging tech
3. **Design Pilots:** Create detailed pilots for high-value applications
4. **Deploy Sandboxes:** Activate specialized sandbox environments for each technology
5. **Ethics & Governance:** Establish frameworks for responsible innovation

Secondary Objectives

- Foster AI literacy across government
- Explore blockchain for transparency and trust
- Understand IoT for smart government
- Build partnerships with tech companies and researchers
- Position T&T as Caribbean emerging tech leader

Target Participants (50-65 people)

Government Stakeholders (20-25)

- **MPAAI Tech Hub Team** (5-6)
- **MDAs with Emerging Tech Potential** (12-15)
- Health (AI diagnostics, patient flow prediction)
- Transportation (IoT traffic management)
- Agriculture (IoT crop monitoring, market prediction)
- Revenue (AI fraud detection)
- Social Services (predictive analytics for interventions)
- Environment (IoT air/water quality, climate monitoring)
- Education (adaptive learning AI)
- Public Safety (predictive policing ethics, emergency response optimization)

- Land registry (blockchain for property records)

- **National-CSIRT** (2-3) - AI security, blockchain security, IoT security

Academic & Research (12-15)

- **Universities** (10-12)
- Computer science faculty (AI/ML, distributed systems specialists)
- Data science researchers
- Graduate students

- Engineering (IoT/embedded systems)

- **Research Institutions** (2-3)

Private Sector & Startups (8-10)

- **AI/ML Companies** (3-4)
- **IoT Companies** (2-3)
- **Blockchain/Crypto Companies** (2-3)

International Experts (8-10)

- **EU AI Ethics Specialist** (1)
- Nordic AI ethics frameworks
- EU AI Act implications
- **AI/ML Technical Expert** (2)
- Government AI applications
- Responsible AI specialist
- **Blockchain Expert** (2)
- Government blockchain use cases (Estonia e-residency, etc.)
- Public vs. permissioned blockchains
- **IoT/Smart City Expert** (2)
- Barcelona/Amsterdam smart city initiatives
- Sensor networks and data analytics
- **CAF & EU Delegation** (2)

Detailed Agenda

Note: This workshop spans 4 days but with 2.5 days of structured sessions plus hands-on lab time

DAY 1: Foundations & AI/ML

Morning Session (9:00 AM - 12:30 PM)

9:00 - 9:30 | Opening & Setting the Stage
- Welcome and workshop objectives
- **Why Emerging Tech Matters for T&T**
- Leapfrogging opportunities
- Competitive advantage
- Better public services
- Economic diversification

9:30 - 11:00 | AI & Machine Learning Fundamentals
- **Presentation:** "AI/ML 101 for Government" (35 min)

- What is AI? What is ML?
- Supervised vs. unsupervised learning
- Deep learning and neural networks (simple explanation)
- Natural Language Processing (NLP)
- Computer vision
- Common applications: Prediction, classification, anomaly detection, recommendation
- Demystifying the hype vs. reality

- **Presentation:** "AI in Government: Global Examples" (35 min)
- **Healthcare:** Diagnostic assistance, patient triage
- **Social Services:** Fraud detection, risk assessment
- **Transport:** Traffic prediction, route optimization
- **Revenue:** Tax fraud detection, audit targeting
- **Citizen Services:** Chatbots, document processing
- **Environmental:** Climate modeling, disaster prediction

- Real examples from Singapore, Estonia, UK, Nordic countries

- **Q&A & Discussion** (20 min)

11:00 - 11:15 | BREAK

11:15 - 12:30 | AI Ethics & Responsible AI
- **Presentation:** "The Dark Side of AI: Bias, Privacy, Control" (30 min)
- Algorithmic bias (racial, gender, socioeconomic)
- Privacy concerns (surveillance, data harvesting)
- Transparency and explainability (black box problem)
- Accountability (who's responsible when AI makes mistakes?)
- Job displacement concerns
- EU AI Act overview

- **Interactive Exercise:** "AI Ethics Scenarios" (40 min)
- Small groups analyze ethical scenarios:
    - **Scenario 1:** AI-assisted hiring for government jobs (bias risk)
    - **Scenario 2:** Predictive policing (fairness, discrimination)
    - **Scenario 3:** Social benefit fraud detection (false positives, vulnerable populations)
    - **Scenario 4:** Automated document processing (errors, accountability)

For each: Identify ethical concerns, propose safeguards

- **Synthesis** (5 min)

12:30 - 1:30 PM | LUNCH

Afternoon Session (1:30 PM - 5:00 PM)

1:30 - 3:00 | AI Use Cases for T&T Government
- **Facilitated Brainstorming:** "Where Can AI Help?" (30 min)
- Participants from each MDA brainstorm AI applications in their domain
- Rapid ideation (quantity over quality)
- Post ideas on walls

- **Use Case Deep Dives** (60 min)
- Breakout groups (by domain):

  - **Group 1: Health & Social Services**
  - **Group 2: Revenue & Finance**
  - **Group 3: Environment & Agriculture**
  - **Group 4: Transport & Infrastructure**
  - **Group 5: Citizen Services & Administration**

For top 2-3 ideas per group, develop:
- Problem statement
- AI approach (what type of ML?)
- Data requirements
- Expected benefits
- Ethical considerations (from WS 1.3)
- Feasibility assessment

- **Gallery Walk** (30 min)
- Groups present use cases
- Dot voting on priorities

3:00 - 3:15 | BREAK

3:15 - 5:00 | Hands-On AI Lab
- **Demo:** "AI in Action" (30 min)
- Live demonstration of ML model:
- Simple classification task (e.g., spam detection)
- Show training data, model training, prediction
- Tools: Python, scikit-learn or similar
- No-code AI tools (Google AutoML, Azure ML)

- **Hands-On Exercise:** "Build Your First ML Model" (60 min)
- Participants use no-code/low-code tools or guided Python notebooks
- Simple government dataset (synthetic)
- Task: Predict outcome (e.g., permit approval time, service request classification)
- Facilitators assist

- Experience the process, not perfection

- **Reflection** (15 min)

- What did you learn?
- How hard/easy was it?
- What skills/tools would government need?

DAY 2: Blockchain & IoT

Morning Session (9:00 AM - 12:30 PM)

9:00 - 10:30 | Blockchain Fundamentals
- **Presentation:** "Blockchain 101: Beyond Cryptocurrency" (40 min)
- What is blockchain? (distributed ledger, immutability)
- How it works (blocks, chains, consensus, cryptography)
- Public vs. private (permissionless vs. permissioned)
- Smart contracts
- Blockchain ≠ Bitcoin (many applications beyond crypto)
- Limitations: Scalability, energy, complexity

- **Presentation:** "Blockchain in Government" (30 min)
- **Land/Property Records:** Georgia, Sweden (blockchain land registries)
- **Digital Identity:** Estonia e-residency
- **Supply Chain:** Provenance tracking (food safety, pharmaceuticals)
- **Voting:** Blockchain voting pilots (security debates)
- **Credentials:** Educational certificates, professional licenses

- **Transparency:** Procurement, budgets, charity tracking

- **Panel Discussion:** "Is Blockchain Right for Government?" (20 min)

- Blockchain expert
- Government IT leader
- Academic researcher
- Hype vs. reality
- When blockchain makes sense vs. traditional database

10:30 - 10:45 | BREAK

10:45 - 12:30 | Blockchain Use Cases & Hands-On
- **Use Case Workshop:** "Blockchain for T&T" (45 min)
- Breakout groups brainstorm blockchain applications:
- **Group 1:** Land registry, property transactions
- **Group 2:** Educational credentials, professional licenses
- **Group 3:** Supply chain (imports, food safety)
- **Group 4:** Transparency (procurement, public spending)

Develop 1-2 use cases per group with:
- Problem being solved
- Why blockchain vs. traditional system?
- Public or private blockchain?
- Participants in the network
- Data to be recorded
- Privacy considerations
- Feasibility and complexity

- **Hands-On Demo:** "Blockchain Sandbox" (45 min)
- Guided exploration of blockchain test network
- Create simple smart contract
- Execute transaction
- View on blockchain explorer
- Tools: Ethereum testnet, Hyperledger, or similar

- Goal: Demystify technology

- **Discussion** (15 min)

- Challenges and opportunities
- Prioritization of use cases

12:30 - 1:30 PM | LUNCH

Afternoon Session (1:30 PM - 5:00 PM)

1:30 - 3:00 | Internet of Things (IoT) Fundamentals
- **Presentation:** "IoT: Connecting the Physical World" (35 min)
- What is IoT? (sensors, connectivity, data, analytics, action)
- Components: Devices, networks, platforms, applications
- Connectivity: WiFi, cellular, LoRaWAN, etc.
- Edge computing vs. cloud
- Examples: Smart homes, wearables, industrial IoT

- **Presentation:** "IoT for Smart Government & Cities" (35 min)
- **Smart City Examples:**
    - Barcelona: Smart parking, waste management, air quality
    - Amsterdam: Smart lighting, traffic management
    - Singapore: Sensors everywhere, data-driven governance
- **Government IoT Applications:**
    - **Environment:** Air/water quality monitoring
    - **Traffic:** Congestion monitoring, adaptive signals
    - **Infrastructure:** Bridge/building structural monitoring

    - **Agriculture:** Soil moisture, weather stations
    - **Energy:** Smart grid, consumption monitoring
    - **Waste:** Smart bins, route optimization
    - **Public Safety:** Emergency detection, crowd monitoring

- **Caribbean Context Discussion** (20 min)

- Climate monitoring (hurricanes, flooding)
- Agriculture (small farmers, precision agriculture)
- Tourism (crowd management, environmental protection)
- Energy (renewable integration, grid resilience)

3:00 - 3:15 | BREAK

3:15 - 5:00 | IoT Use Cases & Security
- **Use Case Workshop:** "IoT for T&T" (50 min)
- Breakout groups design IoT pilots:
- **Group 1:** Smart Traffic & Parking
- **Group 2:** Environmental Monitoring
- **Group 3:** Agriculture & Rural Development
- Group 4: **Energy & Utilities**
- **Group 5:** Public Safety & Emergency Response

For each use case:
- Problem/opportunity
- Sensors/devices needed
- Data collected
- Analytics/insights
- Actions/interventions
- Infrastructure requirements (network, power)
- Cost estimate
- Pilot scope (geographic area, duration)

- **Presentation:** "IoT Security & Privacy" (25 min)
- IoT security challenges:

    - Device vulnerabilities (weak passwords, no updates)
    - Network attacks (interception, DDoS)
    - Data privacy (constant surveillance)

- Best practices:

    - Security by design
    - Encryption
    - Regular updates
    - Network segmentation

- Privacy considerations:
    - Data minimization
    - Purpose limitation
    - Consent (if personal data)
- **Report Back** (20 min)
- Groups present IoT use cases
- Security/privacy review

DAY 3: Integration, Pilots & Strategy

Morning Session (9:00 AM - 12:30 PM)

9:00 - 10:00 | Convergence: AI + Blockchain + IoT
- **Presentation:** "Emerging Tech Convergence" (30 min)
- How technologies combine:
- **IoT + AI:** Sensor data → ML analytics → Automated responses
- **Blockchain + AI:** Transparent AI decisions, data provenance
- **Blockchain + IoT:** Secure device identity, immutable sensor records
- **All Three:** Smart contracts triggered by IoT data analyzed by AI
- Example: Smart agriculture
    - IoT sensors collect soil/weather data
    - AI predicts optimal irrigation/fertilization
    - Blockchain records supply chain from farm to market
- **Interactive Discussion:** "Convergence Use Cases" (30 min)
- Revisit use cases from Days 1-2
- Which could benefit from technology integration?
- How would integration add value?

10:00 - 10:15 | BREAK

10:15 - 12:30 | Pilot Selection & Design
- **Prioritization Exercise:** "Selecting Pilots" (40 min)
- All use cases from Days 1-2 on display
- Criteria:
- Impact (high value to citizens/government)
- Feasibility (technical, financial, timeline)
- Learning (builds capacity, demonstrable)

- Scalability (can expand if successful)
- Equity (serves all populations, not just privileged - WS 1.3)

- Facilitated discussion and voting
- Select 3-4 pilots (ideally 1-2 from each technology area)
- **Pilot Design Workshop** (95 min)
- Break into pilot groups
- Design detailed pilot plans (similar to WS 2.3 approach):
  - Objectives and success metrics
  - Technical approach
  - Data requirements and sources
  - Infrastructure and tools
  - Team and skills needed
  - Timeline (6-9 month pilot)
  - Budget estimate
  - Risks and mitigation
  - Ethical/privacy safeguards
  - User testing and evaluation plan

12:30 - 1:30 PM | LUNCH

Afternoon Session (1:30 PM - 4:30 PM)

1:30 - 2:45 | Emerging Tech Governance & Strategy
- **Presentation:** "Governing Emerging Technologies" (25 min)
- Policy frameworks needed:
- AI ethics policy (building on discussions from Day 1)
- Data governance for AI/ML (training data, model outputs)
- Blockchain governance (who controls the network?)
- IoT security standards
- Procurement considerations
- Skills and capacity building
- Research and innovation funding

- **Workshop:** "T&T Emerging Tech Strategy" (45 min)
- Collaborative development of strategy framework:
  - **Vision:** T&T's emerging tech ambitions (national, regional)
  - **Principles:** Ethical, inclusive, secure, open
  - **Focus Areas:** Priority technologies and applications
  - **Enablers:** Infrastructure (sandboxes), skills, partnerships
  - **Governance:** Oversight, standards, accountability

    - **Roadmap:** Phases (learn → pilot → scale)

- Draft outline of Emerging Tech Strategy document

2:45 - 3:00 | BREAK

3:00 - 4:00 | Sandbox Deployment & Community Building
- **Technical Presentation:** "Emerging Tech Sandboxes" (20 min)
- Specialized sandboxes in tech hub (from WS 3.1):
- **AI/ML Sandbox:** GPUs, ML frameworks (TensorFlow, PyTorch), Jupyter notebooks, datasets
- **Blockchain Sandbox:** Test networks (Ethereum, Hyperledger), smart contract dev tools
- **IoT Sandbox:** Device simulators, sensor data streams, edge computing platforms

- Access procedures and support

- **Community & Skills Development** (35 min)

- Training programs needed:

    - Government staff AI literacy
    - Data science bootcamps
    - Blockchain development courses
    - IoT engineering workshops

- University partnerships for research and talent
- Industry partnerships for expertise and tools
- Regional knowledge exchange (build on WS 2.2)
- Hackathons and challenges specific to emerging tech

4:00 - 4:30 | Day 3 Wrap & Free Lab Time Preview
- Summary of pilots and strategy
- **Preview of Day 4:** Optional hands-on lab time (half day)

DAY 4 (OPTIONAL): Hands-On Lab Time (9:00 AM - 1:00 PM)

Purpose: Open sandbox time for participants to experiment, with expert support available

Structure:
- **Self-directed exploration** in AI/ML, Blockchain, or IoT sandboxes
- **Expert office hours** - rotating availability from technical experts
- **Peer collaboration** - participants work together on ideas
- **Project pitches** (optional) - participants pitch sandbox project ideas for potential support
- **Wrap-up** (12:30-1:00 PM) - reflections, feedback, next steps

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Key Discussion Questions

Strategic Questions

- Which emerging technologies offer the most value for T&T context?
- How do we balance innovation with responsible governance?
- What investments in infrastructure and skills are justified?
- How can T&T position as regional emerging tech leader?

Ethical Questions

- How do we ensure AI doesn't perpetuate or amplify bias?
- Who is accountable when AI makes wrong decisions?
- How do we protect privacy in an IoT-enabled world?
- How do we ensure emerging tech benefits all, not just privileged?

Technical Questions

- What infrastructure is needed for AI/ML at scale?
- Public or private blockchain for government use?
- How do we secure IoT devices and networks?
- What skills gaps must be filled?

Practical Questions

- Are we trying to do too much too fast?
- How do we learn from failures without wasting resources?
- How do we sustain pilots beyond initial funding?
- How do we scale successful pilots?

Expected Outputs & Deliverables

Immediate Outputs (During Workshop)

1. **Prioritized use case catalogue** (15-20 use cases across AI/ML, blockchain, IoT)
2. **Detailed pilot designs** (3-4 pilots ready for implementation)
3. **Emerging Tech Ethics Framework** (principles and safeguards)
4. **Sandbox specifications** for AI/ML, blockchain, IoT
5. **Emerging Tech Strategy** (draft framework)
6. **Skills development plan**

Post-Workshop Deliverables (2-3 weeks)

1. **Workshop Report** (Product 3)
2. **Emerging Tech Strategy Document** (Appendix)
3. **Pilot Implementation Plans** (detailed specs for each pilot)
4. **Training & Capacity Building Plan** (Appendix)

Success Metrics

Participation: 50+ participants, diverse sectors represented

Engagement: 15+ use cases identified, 3-4 pilots designed

Satisfaction: 4.5/5 average, 90%+ understand technologies better

Outcome: Pilots begin implementation within 1 month, sandboxes deployed

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Connection to Other Workshops

Builds On: WS 1.2 (infrastructure), WS 1.3 (ethics), WS 3.1 (sandboxes)
Builds For: WS 3.3 (private sector pilots), WS 3.4/4.1 (showcase results), WS 4.2 (governance)

Links to TOR Requirements

TOR Section 5, Objective C:
- **Action c:** "Support the conduct of workshops (approx. 5) to develop the data centre as a tech hub with sandboxes for open-source, Artificial Intelligence (AI), Machine Learning (ML), blockchain, and Internet of Things (IoT)." [This workshop covers workshops #2-5]

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